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THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



CVE-2026-42012 Gnutls: gnutls: certificate validation bypass due to improper handling of uri and srv sans

Vulnerability Report

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CVE-2026-42012 GnuTLS certificate validation bypass via URI and SRV SAN handling

1. Executive Summary

CVE-2026-42012 is a certificate-validation bypass in GnuTLS. Upstream GnuTLS advisory material, the public GitLab issue, NVD, Ubuntu, Debian, and downstream Linux-security tracking all converge on the same core flaw: when a certificate contains Uniform Resource Identifier (URI) or Service (SRV) Subject Alternative Names (SANs), vulnerable GnuTLS hostname-validation logic can incorrectly fall back to comparing the requested DNS hostname against the certificate Common Name (CN). That fallback should not occur once SAN data is present.

The practical security consequence is service-identity spoofing risk rather than memory corruption or direct remote code execution. Under the most credible attack scenario described in the upstream issue, an attacker who can obtain or abuse a certificate chain trusted by the victim environment may present a crafted certificate containing a URI or SRV SAN plus a victim hostname in the CN field. A GnuTLS-using application could then accept the certificate for the wrong purpose, enabling impersonation of an internal or external service and creating realistic man-in-the-middle or traffic-interception conditions.

Public reporting reviewed for this task did not show confirmed in-the-wild exploitation, CISA KEV inclusion, or widely circulated weaponized proof-of-concept tooling as of 2026-06-11. Even so, this issue matters for enterprises that use GnuTLS-backed TLS validation in internal PKI, service-discovery, mTLS, or private-CA environments where certificate profile misuse is plausible. The highest-confidence remediation is to upgrade upstream GnuTLS to `3.8.13` or later, or apply vendor backports such as the Debian and Ubuntu fixed package releases published after the upstream advisory.

Executive takeaway	Assessment
Business impact	Incorrect certificate acceptance can let an attacker spoof a trusted service or intercept sensitive traffic when they can present a crafted certificate in a trust path the victim accepts.
Most exposed environments	GnuTLS-based clients, proxies, or services performing hostname validation in enterprise/private-PKI environments, especially where URI or SRV SAN-bearing certificates may exist.
Exploitation prerequisites	The attacker must influence the certificate presented to the victim and typically needs a trusted issuance path, compromised CA path, or equivalent certificate-positioning advantage.
Immediate priority	Upgrade to upstream `3.8.13+` or vendor-fixed backports, then review internal certificate-validation dependencies and CA issuance practices.

2. Intelligence Requirement

This review addresses the following intelligence requirement: what does current public reporting say about `CVE-2026-42012`, which GnuTLS deployment patterns are actually exposed, how realistic is exploitation in enterprise environments, and what concrete actions should defenders, PKI owners, and detection teams take next?

Question	Assessment
What is the vulnerability?	A certificate-validation bypass in GnuTLS where URI or SRV SAN-bearing certificates can still trigger DNS hostname checks against the CN, contrary to expected SAN-first validation behavior.
What is affected?	GnuTLS implementations performing certificate identity checks

	with vulnerable releases or downstream package versions lacking the upstream fix.
What is the attacker outcome?	Potential spoofing of legitimate services or interception of sensitive traffic by getting a victim to trust a certificate it should have rejected.
Is this a generic TLS break for every stack?	No. The issue is specific to GnuTLS validation behavior; OpenSSL-, Schannel-, or NSS-backed applications are not affected merely because they use TLS.
Is exploitation publicly confirmed?	No confirmed in-the-wild exploitation was identified in reviewed public sources as of 2026-06-11.
What matters most operationally?	Patch vulnerable GnuTLS deployments, identify where GnuTLS performs hostname checks, and review PKI issuance and trust assumptions around URI/SRV SAN certificates.

3. Source Review and Confidence

Source	Contribution	Confidence
Microsoft Security Update Guide intake entry	Confirms the intake event, CVE title, and publication chronology used to trigger this report.	Medium
GnuTLS security advisory `GNUTLS-SA-2026-04-29-7`	Primary upstream statement of the flaw, severity, root issue summary, reporter attribution, and recommendation to upgrade to `3.8.13` or later.	High
GnuTLS 3.8.13 release announcement	Corroborates the fix narrative and states that URI/SRV SANs should suppress CN fallback.	High
GitLab issue `#1802`	Richest public technical source for root cause, impact conditions, and the PKI misuse scenario: URI/SRV SAN presence did not preclude CN fallback.	High
Upstream fix commit `8dcc6a1f48945997666ac9f10896819edd01a03b`	Canonical remediation reference showing the hostname-verification logic change that makes URI/SRV SANs preclude CN fallback.	High
NVD entry	Provides broad public description and `CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:H/A:N` scoring context.	High
Ubuntu CVE tracker / USN lineage	Confirms Canonical shipped fixed backports for maintained releases and links the fix to upstream patch `8dcc6a1`.	High
Debian security tracker	Useful downstream packaging view of vulnerable and fixed versions across bullseye, bookworm, trixie, and unstable.	High
CISA weekly vulnerability bulletin (`sb26-	Corroborates public-sector awareness but	Medium

152`)	does not add exploitation confirmation.	
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Confidence is high on the existence, root cause, and remediation of the vulnerability because upstream GnuTLS material, the public issue discussion, and multiple downstream package trackers align closely on what went wrong and how it was fixed. Confidence is also high that the issue affects identity validation semantics rather than memory safety or direct code execution.

Confidence is only medium on broad enterprise exploitation likelihood because the reviewed public material does not show widespread exploitation, exploit-kit adoption, or field telemetry. The issue becomes materially more serious in environments where an attacker can influence certificate issuance, internal trust anchors, private PKI, or service-discovery certificate profiles.

4.1 Core Vulnerability Metadata

Field	Value
CVE	`CVE-2026-42012`
Product	`GnuTLS`
Component	`libgnutls` certificate hostname verification
Upstream advisory	`GNUTLS-SA-2026-04-29-7`
Vulnerability class	Certificate validation bypass / service spoofing risk
Public upstream severity	`Medium` per GnuTLS advisory
NVD CVSS v3.1	`7.1 / HIGH`
NVD vector	`CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:H/A:N`
Vulnerable behavior	URI or SRV SAN-bearing certificates could still fall back to DNS-hostname comparison against the Common Name
Security consequence	Acceptance of a certificate beyond its intended purpose, enabling spoofing or interception opportunities
Upstream fixed version	`3.8.13`
Canonical fix commit	`8dcc6a1f48945997666ac9f10896819edd01a03b`
Debian fixed examples	`3.7.1-5+deb11u10` (bullseye security), `3.7.9-2+deb12u7` (bookworm security), `3.8.9-3+deb13u4` (trixie security), `3.8.13-1` (unstable)
Ubuntu fixed examples	`3.8.12-2ubuntu1.1` (26.04), `3.8.9-3ubuntu2.2` (25.10), `3.8.3-1.1ubuntu3.6` (24.04), `3.7.3-4ubuntu1.9` (22.04)
Public exploitation at review time	Not confirmed in reviewed sources
KEV status at review time	Not listed in reviewed sources

4.2 Root Cause and Validation Failure

The public GitLab issue and upstream release notes describe a standards-relevant validation flaw: URI and SRV SAN entries were not treated as sufficient SAN presence to suppress CN fallback. In practical terms, the validator behaved as though only certain SAN types mattered for deciding whether hostname matching should ignore the CN field.

Root-cause element	Analyst assessment
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Validation decision point	Vulnerable logic allowed CN fallback even when URI or SRV SANs were present in the certificate.
Why this is dangerous	RFC-aligned SAN handling exists to prevent certificates from being accepted for identities they were not meant to represent.
Upstream technical clue	Issue `#1802` states that only DNS and IP SAN types were treated as supported in the relevant decision path, so URI/other SAN types did not block the CN fallback branch.
Fix behavior	Commit `8dcc6a1f...` changes hostname verification so URI/SRV SANs preclude CN fallback.
Security lesson	“Present but ignored” identity metadata can be as dangerous as a total absence of validation if the fallback path is overly permissive.

4.3 Exposure and Affected Deployment Patterns

Exposure pattern	Why it matters	Confidence
GnuTLS-backed applications validating remote service certificates	These are the direct consumers of the flawed hostname-verification path.	High
Enterprise/private-PKI environments	Internal trust anchors make it more plausible that a malicious or compromised issuance path could create a crafted certificate that victims still trust.	Medium
Service-discovery or non-DNS certificate profiles using URI/SRV SANs	The flaw is specifically about URI/SRV SAN handling and misuse of fallback behavior.	High
TLS-inspecting gateways, proxies, or internal middleware built on GnuTLS	These systems can magnify impact because they sit in front of many downstream users or services.	Medium
Non-GnuTLS TLS stacks	The reviewed public evidence does not support claiming exposure merely because the application uses TLS generally.	High

4.4 Realistic Attack Scenarios

Scenario	Description	Defensive implication
Private-CA service spoofing	An attacker with access to a trusted private CA, compromised intermediate, or weak issuance workflow obtains a certificate containing a URI SAN plus a victim hostname in CN. Vulnerable GnuTLS clients trust it for the hostname request.	Review issuance controls and trust-anchor governance, not just package versions.
Internal mTLS or service-mesh misuse	A crafted certificate intended for one service identity is accepted in another hostname-validation context because CN fallback occurs despite SAN presence.	Validate certificate profiles and test critical internal trust paths after patching.

Targeted interception in controlled PKI	An adversary able to place a crafted certificate in a trusted path uses the validation mismatch to impersonate a service and capture credentials or application traffic.	Treat this as a realistic man-in-the-middle enabler where trust-path compromise is plausible.
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4.5 Analyst Findings

Finding	Defensive implication
This is a trust-boundary and identity-validation flaw, not an availability or code-execution bug.	Prioritize service-identity assurance, PKI governance, and dependency patching over exploit-signature hunting alone.
The highest-value risk appears in managed enterprise PKI or internal-service ecosystems.	Exposure management should focus on where GnuTLS validates identities inside the organization, not only on internet-facing servers.
Public reporting supports a narrowly described but meaningful exploitation prerequisite set.	Avoid sensational claims of universal spoofing while still treating trusted-certificate abuse scenarios as serious.
Vendor backports are available across major Linux distributions.	Defenders do not necessarily need to wait for upstream-only packaging; distribution fixes may already be sufficient when verified.

5. Threat Context

CVE-2026-42012 sits in the class of certificate-validation failures that become dangerous when organizations over-trust internal PKI or assume every SAN-bearing certificate is being interpreted in a standards-compliant way. The upstream advisory language is measured, but the GitLab issue makes the enterprise impact clearer: controlled PKI deployments can mis-validate server identity if a malicious or compromised issuance path creates a certificate with URI SAN data while also placing the victim hostname in the CN.

That means this vulnerability is not most likely to be exploited as indiscriminate internet noise. Instead, it is better understood as an adversary-enablement issue for targeted interception, service impersonation, or abuse of weak certificate-governance processes. In mature environments, attackers increasingly target identity systems and trust relationships because they provide stealthier access than overt exploitation of a memory-corruption bug. A validation bypass in a widely deployed TLS library fits that pattern.

Reviewed public sources did not identify a named threat actor, proof of in-the-wild abuse, or a public exploit campaign linked to this CVE as of 2026-06-11. The absence of exploitation reporting should not be treated as proof of low impact. In environments where internal services, private trust anchors, or service-mesh identities matter operationally, a standards-edge validation flaw can be a high-value enabler even without commodity exploitation.

MITRE ATT&CK mapping	Relevance	Confidence
`T1557` Adversary-in-the-Middle	The vulnerability can help an attacker impersonate a service and intercept traffic when they can present a crafted certificate in a trusted path.	Medium
`T1199` Trusted Relationship	Low-confidence but relevant where abuse of internal PKI or trust anchors lets an attacker exploit the validation gap across trusted service relationships.	Low
Additional ATT&CK techniques	Not assigned with high confidence because public reporting does not describe a full	High

	intrusion chain beyond certificate misuse/spoofing potential.	
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6. Detection and Hunting

The strongest detection strategy is exposure-centric rather than IOC-centric. Public reporting reviewed for this task did not provide durable file hashes, domains, or exploit artifacts. Defenders will get more value from identifying vulnerable GnuTLS versions, locating applications that depend on GnuTLS for hostname validation, and reviewing certificate inventories for URI/SRV SAN-bearing profiles that could be misused.

6.1 Detection Logic Summary

Signal	Telemetry source	Analytic goal
GnuTLS versions below upstream `3.8.13` or below vendor-fixed package builds	SBOM platforms, package inventory, vulnerability management, container image scanning	Identify systems requiring immediate remediation
Certificates with URI or SRV SAN entries plus hostname-like CN values	Internal PKI inventory, certificate-management platforms, issuance logs, CT-style internal transparency stores	Find certificate profiles that could be misinterpreted by vulnerable validators
GnuTLS-linked applications handling outbound TLS hostname verification	Application dependency inventories, build manifests, package ownership, linked-library analysis	Prioritize systems where the bug is actually reachable
Unexpected service-identity anomalies after certificate rotation or internal CA events	Reverse proxies, mTLS gateways, service mesh telemetry, auth logs	Surface possible certificate misuse or trust-path drift
Distribution advisories or scanner findings explicitly tracking `CVE-2026-42012`	Vulnerability scanners, package-manager advisories, EDR vulnerability modules	Confirm remediation completion and residual exposure

Analyst caveat: package presence alone is not final proof of a critical exploit path. The highest-confidence risk exists where GnuTLS is actively performing hostname checks for sensitive service identities.

6.2 Example Detection / Hunt Content

Microsoft Defender vulnerability inventory query:

```
DeviceTvmSoftwareInventoryVulnerabilities
| where CveId == "CVE-2026-42012"
| project DeviceName, OSPlatform, SoftwareName, SoftwareVersion, VulnerabilitySeverityLevel,
RecommendedSecurityUpdate
| order by DeviceName asc
```

Linux package-inventory triage for common distributions:

```
printf 'Debian/Ubuntu package view:\n'; dpkg-query -W 'libgnutls30*' 2>/dev/null || true
printf '\nRPM package view:\n'; rpm -qa | grep -Ei '^gnutls|^libgnutls' || true
```

Certificate-inventory hunt for URI/SRV SAN-bearing certificates with CN reuse risk:

```
find /etc/ssl /usr/local/share/ca-certificates /opt -type f \( -name '*.crt' -o -name '*.pem' \)
2>/dev/null \
| while read -r cert; do
  openssl x509 -in "$cert" -noout -subject -ext subjectAltName 2>/dev/null \
  | awk -v file="$cert" '
    /subject=/ { subj=$0 }
    /URI:|SRVName:/ { print "FILE=" file "\n" subj "\n" $0 "\n" }
  '
done
```

Sigma-style review stub for PKI and certificate-inventory analysis workflows:


```

title: Review URI or SRV SAN Certificates in GnuTLS-Dependent Environments
id: cve-2026-42012-uri-srv-san-review
status: experimental
description: Identifies certificate profiles and PKI workflows that should be reviewed for CVE-2026-42012 exposure.
logsource:
  product: pki
  category: certificate_inventory
detection:
  selection_san:
    SubjectAlternativeName|contains:
      - 'URI:'
      - 'SRVName:'
  selection_cn:
    Subject|contains: 'CN='
  condition: selection_san and selection_cn
level: medium

```

6.3 Hunting Guidance

Hunt question	Why it matters
Which assets still carry GnuTLS package versions flagged for `CVE-2026-42012`?	This defines the immediate patch set.
Which business-critical applications actually rely on GnuTLS for hostname verification?	Reachability matters more than package presence alone.
Where does the organization issue or trust URI/SRV SAN-bearing certificates?	Those certificate profiles are closest to the vulnerable logic branch.
Do internal CA, service-mesh, or proxy logs show unusual certificate substitutions, validation failures, or certificate-profile drift?	This can reveal unsafe trust relationships or attempted misuse.
Have systems been patched but not restarted, leaving old GnuTLS-linked processes resident?	Library-based fixes may not take full effect until dependent services reload.

7. Recommendations

Priority	Owner	Action
Critical	Platform Engineering / System Owners	Upgrade upstream GnuTLS to `3.8.13` or later, or deploy vendor backports confirmed to remediate `CVE-2026-42012`.
Critical	Vulnerability Management	Inventory all packages, containers, and appliances carrying vulnerable GnuTLS builds and track remediation to closure.
High	PKI / Identity Engineering	Review issuance policy for certificates containing URI or SRV SANs and confirm they cannot be misused across service-identity boundaries.
High	Application Security	Identify applications that depend on GnuTLS hostname verification and validate certificate acceptance behavior after patching.
High	Detection Engineering	Add package/SBOM detection for

		`CVE-2026-42012` and certificate-inventory hunts for URI/SRV SAN-bearing profiles in sensitive environments.
Medium	Infrastructure / Service Owners	Restart or redeploy long-lived services after patching so they load the fixed library version.
Medium	Security Architecture	Review whether internal trust anchors, service-mesh certs, or private-CA workflows create high-value certificate-spoofing opportunities.
Medium	Threat Intelligence	Monitor future GnuTLS, distro, NVD, and CISA updates for any change in exploitation status or additional technical detail.

Additional analyst guidance:

- Treat this as both a dependency-patching issue and a trust-model review. The vulnerability matters most where certificate semantics are an operational control, not just a background library feature.
- Do not assume all “fixed” environments are equal until package versions and service restarts are verified on the assets that actually use GnuTLS.
- Where private PKI is heavily used, include certificate-profile review in remediation, not only package updates.

8. References

- 1. Microsoft Security Update Guide - CVE-2026-42012: <https://msrc.microsoft.com/update-guide/vulnerability/CVE-2026-42012>
- 2. GnuTLS Security Advisory `GNUTLS-SA-2026-04-29-7`: <https://www.gnutls.org/security-new.html#GNUTLS-SA-2026-04-29-7>
- 3. GnuTLS 3.8.13 release announcement: <https://lists.gnutls.org/pipermail/gnutls-help/2026-April/004922.html>
- 4. GitLab issue `#1802` - `gnutls CN fallback with unsupported SAN type`: <https://gitlab.com/gnutls/gnutls/-/issues/1802>
- 5. Upstream fix commit `8dcc6a1f48945997666ac9f10896819edd01a03b`: <https://gitlab.com/gnutls/gnutls/-/commit/8dcc6a1f48945997666ac9f10896819edd01a03b>
- 6. NVD entry - CVE-2026-42012: <https://nvd.nist.gov/vuln/detail/CVE-2026-42012>
- 7. Ubuntu CVE tracker - CVE-2026-42012: <https://ubuntu.com/security/CVE-2026-42012>
- 8. Debian security tracker - CVE-2026-42012: <https://security-tracker.debian.org/tracker/CVE-2026-42012>
- 9. CISA vulnerability summary bulletin `sb26-152`: <https://www.cisa.gov/news-events/bulletins/sb26-152>